

# Triangles Touching Three Sides of a Square

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30 September 2026

doi:10.5281/zenodo.23057791

## Abstract

We study a boundary-contact variant of the classical problem of placing a triangle in a square. A triangle has *three-side contact* if its three vertices can be assigned to three distinct sides of the square. For a fixed nondegenerate triangle  $T$ , we prove that the admissible square side lengths form one interval  $[m(T), c]$ , where  $c$  is the longest side of  $T$ . After normalizing  $c = 1$ , we give an explicit four-region formula for the lower endpoint  $m(T)$  and explain geometrically why this differs from ordinary containment in exactly one region. As an arithmetic application, let  $a(n)$  count congruence classes of integer-sided triangles having three-side contact with a square of side  $n$ . We obtain an exact finite test and

$$a(n) = Cn^3 + O(n^2),$$

where  $C = 0.0776924549\dots$  is given by an explicit area integral.

## 1 Why three-side contact?

A classical question asks for the smallest square that contains a given triangle. The complete answer is known; see Jerrard and Wetzel [2] and Yuan and Ding [3]. Here we impose one extra condition: the three triangle vertices must touch three distinct sides of the square. A corner belongs to two sides geometrically, but a triangle vertex at a corner is assigned to only one of them.

The extra condition isolates a feature that ordinary containment does not force: support by three triangle vertices. A smallest containing square may have one triangle side as a diagonal while the third vertex remains strictly inside the square. Requiring three-side contact removes exactly this possibility. The simplest example is the isosceles triangle with side lengths  $(2, 2, 3)$ .

Two questions then arise before any counting problem is considered.

First, if the size of the square is varied while the triangle is kept fixed, can admissible sizes appear and disappear, or do they form one interval? Second, can the smallest admissible square be read directly from the shape of the triangle? Theorem 3 answers the first question and Theorem 6 the second.

The integer problem is then a discrete application of this geometry. For a positive integer  $n$ , let  $a(n)$  be the number of congruence classes of triangles with positive integer side lengths having three-side contact with a square of side  $n$ . The first terms are

$$1, 2, 6, 10, 19, 28, 42, 58, 82, 108, 143, 176, 223, 271, 331, 397, \dots,$$

and form OEIS A400509.

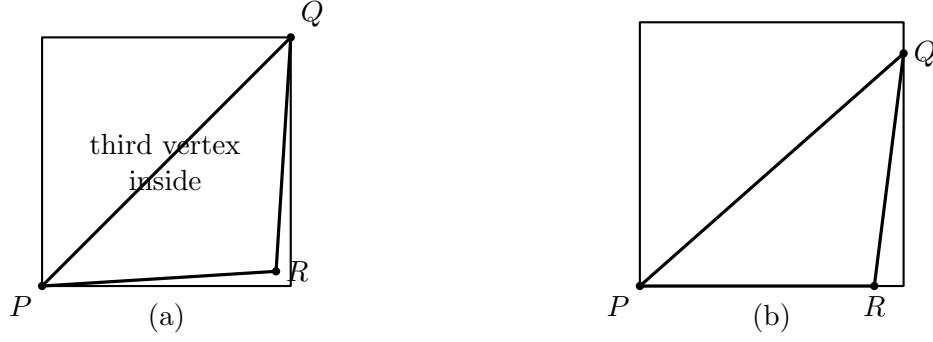


Figure 1: The triangle  $(2, 2, 3)$  illustrates the difference between (a) ordinary containment, where a smallest square has side  $3/\sqrt{2}$  and leaves the third vertex in the interior, and (b) three-side contact, whose smallest square has side  $9/4$ .

## 2 The contact geometry

Let a triangle have vertices  $P, Q, R$ , with

$$PQ = q, \quad PR = r, \quad QR = s.$$

We first prescribe that  $P$  and  $Q$  lie on opposite sides of a square of side  $n$ , while  $R$  lies on a third side. This is the basic contact type; there are only six ordered choices of  $(q, r, s)$  for a given triangle.

Put

$$H = \sqrt{(q + r + s)(-q + r + s)(q - r + s)(q + r - s)}.$$

Thus  $H = 4\Delta$ , where  $\Delta$  is the area of the triangle. Also put

$$U = q^2 + r^2 - s^2, \quad V = q^2 + s^2 - r^2,$$

and, when  $q \geq n$ ,

$$t = \frac{\sqrt{q^2 - n^2}}{n}.$$

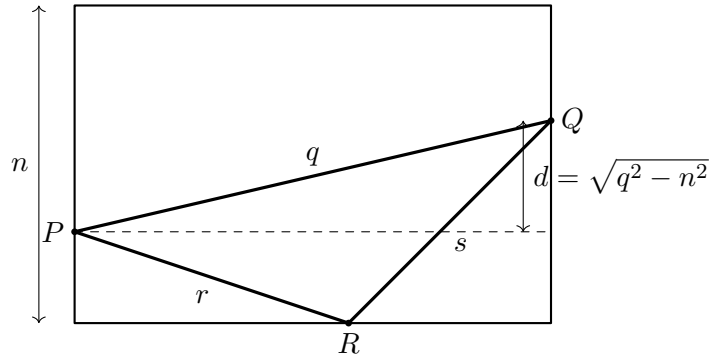


Figure 2: One contact type. The side  $PQ = q$  joins two opposite sides of the square; the third vertex lies on a third side. Once  $q, r, s$  and  $n$  are fixed, the placement is determined up to reflections.

**Theorem 1** (Exact contact test). *The prescribed contact type exists if and only if*

$$n \leq q \leq \sqrt{2}n$$

and

$$0 \leq U + Ht \leq 2q^2, \quad (1)$$

$$0 \leq H - Ut \leq 2q^2, \quad (2)$$

$$0 \leq H + Vt \leq 2q^2. \quad (3)$$

*Proof.* Write

$$P = (0, \eta), \quad Q = (n, \eta + d), \quad R = (\xi, 0),$$

where  $d = \sqrt{q^2 - n^2}$ . The projection of  $PR$  on  $PQ$  has length  $U/(2q)$ , while the altitude over  $PQ$  has length  $H/(2q)$ . Resolving  $PR$  into directions parallel and perpendicular to  $PQ$  gives

$$\xi = \frac{nU + dH}{2q^2}, \quad \eta = \frac{nH - dU}{2q^2}.$$

Since  $t = d/n$ ,

$$\frac{\xi}{n} = \frac{U + Ht}{2q^2}, \quad \frac{\eta}{n} = \frac{H - Ut}{2q^2}.$$

Using  $U + V = 2q^2$  also gives

$$\frac{\eta + d}{n} = \frac{H + Vt}{2q^2}.$$

Thus (1)–(3) say precisely that  $R$ ,  $P$ , and  $Q$  lie on the required three sides of the square. Finally, the distance between points on opposite sides of an  $n \times n$  square lies between  $n$  and  $\sqrt{2}n$ .  $\square$

The theorem replaces a continuous search over rotations and boundary points by six elementary algebraic tests. For example, for the 3, 4, 5 triangle the smallest admissible square has side

$$\frac{16}{\sqrt{17}} = 3.880570\dots;$$

see Example 5 below.

### 3 Admissible square sizes have no gaps

We now fix the triangle and vary the size of the square. The key point is that a contact type cannot simply end at its smallest scale: at such an endpoint, the same placement can be reinterpreted using a longer triangle side across the square.

Scale the square to side 1. For a chosen opposite-side length  $q$ , define

$$\mu = \frac{q^2 + r^2 - s^2}{2q^2}, \quad \omega = \frac{2\Delta}{q^2}.$$

If the scaled length of  $q$  is  $\sqrt{1+t^2}$ , one may write, after reflection if necessary,

$$P = (0, \omega - \mu t), \quad Q = (1, \omega + (1 - \mu)t), \quad R = (\mu + \omega t, 0).$$

Hence this contact type is possible exactly when

$$0 \leq \mu + \omega t \leq 1, \quad 0 \leq \omega - \mu t \leq 1, \quad 0 \leq \omega + (1 - \mu)t \leq 1. \quad (4)$$

For one fixed contact type these are six linear inequalities in  $t$ , so the admissible values of  $t$  form an interval.

**Lemma 2** (Endpoint handoff). *If the left endpoint of a nonempty contact interval is realized with a triangle side that is not longest, then the same square admits a contact type in which a strictly longer side joins two opposite sides.*

*Proof.* There are two cases.

Suppose first that the endpoint has  $t = 0$ . Then

$$P = (0, \omega), \quad Q = (1, \omega), \quad R = (\mu, 0),$$

with  $0 \leq \mu \leq 1$  and  $0 < \omega \leq 1$ . If  $PQ$  is not longest, assume after reflection that

$$p = PR = \sqrt{\mu^2 + \omega^2} > 1.$$

Set  $k = \sqrt{p^2 - 1}$  and

$$c_0 = \frac{\mu - \omega k}{p^2}, \quad s_0 = \frac{\omega + \mu k}{p^2}.$$

A rotation sending  $(\mu, -\omega)$  to  $(1, k)$  sends  $(1, 0)$  to  $(c_0, s_0)$ . The inequalities  $\mu, \omega \leq 1$  imply

$$0 \leq c_0 \leq 1, \quad k \leq s_0 \leq 1.$$

Indeed,  $k \leq \mu, \omega$  gives  $c_0 \geq 0$ ; moreover

$$(p^2 - \omega)^2 - \mu^2 k^2 = p^2(\omega - 1)^2 \geq 0$$

gives  $s_0 \leq 1$ , while  $k(p^2 - \mu) \leq \omega$  gives  $s_0 \geq k$ . After translation the rotated triangle has vertices

$$P' = (0, 1 - s_0), \quad R' = (1, 1 - s_0 + k), \quad Q' = (c_0, 1).$$

Thus the longer side  $P'R'$  joins opposite sides of the same unit square.

Now suppose the left endpoint has  $t > 0$ . If  $\mu \geq 0$ , then  $t = 0$  would also satisfy (4), so necessarily  $\mu < 0$ . The left endpoint is therefore  $t = -\mu/\omega$ , where  $\mu + \omega t = 0$ . Write the resulting placement as

$$R = (0, 0), \quad P = (0, Y), \quad Q = (1, Z),$$

with  $0 < Y < Z \leq 1$ . Then

$$PQ = \sqrt{1 + (Z - Y)^2}, \quad RQ = \sqrt{1 + Z^2} > PQ.$$

Rotate the vector  $(1, Z)$  to  $(1, -Z)$ . Under this rotation  $(0, Y)$  becomes  $(\xi, \eta)$ , where

$$\xi = \frac{2YZ}{1 + Z^2}, \quad \eta = \frac{Y(1 - Z^2)}{1 + Z^2}.$$

Since  $0 < Y < Z \leq 1$ , one has  $0 \leq \xi \leq 1$  and  $\eta + Z \leq 1$ . After translation, the same unit square therefore has a contact type in which the strictly longer side  $RQ$  joins its opposite sides.  $\square$

**Theorem 3** (No gaps). *Let  $T$  be a nondegenerate triangle and let  $c$  be its longest side. There is a number  $m(T)$  such that  $T$  has three-side contact with a square of side  $n$  exactly when*

$$m(T) \leq n \leq c.$$

*Proof.* For each of the six contact types, (4) shows that the admissible scale factors form an interval. By Lemma 2, the left endpoint of any interval based on a non-longest side belongs to an overlapping interval based on a strictly longer side. After at most two handoffs one reaches a longest side.

For a longest side  $c$ , the value  $t = 0$  is always possible: the projection parameter lies in  $[0, 1]$ , and the altitude to a longest side is at most  $c$ . Hence all contact intervals belong to one connected chain ending at the scale for which the square side is  $c$ . Returning from scale factors to square side lengths gives one interval  $[m(T), c]$ .  $\square$

The theorem has a simple interpretation: once a fixed triangle fits with three-side contact, enlarging the square does not create a later obstruction before the square side reaches the longest side of the triangle.

*Example 4.* For an equilateral triangle of side  $k$ ,

$$m(T) = k \cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4} k.$$

Thus its admissible square sizes are exactly

$$k \cos 15^\circ \leq n \leq k.$$

*Example 5.* For a right triangle with legs  $p \leq q$ ,

$$m(T) = \frac{q^2}{\sqrt{q^2 + (q - p)^2}}.$$

For the 3, 4, 5 triangle this gives

$$m(T) = \frac{16}{\sqrt{17}}.$$

## 4 The smallest square from the shape of the triangle

The previous theorem reduces the problem to the lower endpoint  $m(T)$ . Normalize the longest side to 1 and write the side lengths as

$$(x, y, 1), \quad 0 < x \leq y \leq 1, \quad x + y > 1.$$

Let  $\rho(x, y)$  be the largest factor by which this normalized triangle can be enlarged while retaining three-side contact with the unit square. Then

$$m(x, y, 1) = \frac{1}{\rho(x, y)}.$$

For this normalized triangle, the quantities  $H, U, V$  from Section 2 become

$$H = \sqrt{4x^2y^2 - (x^2 + y^2 - 1)^2},$$

$$U = 1 + x^2 - y^2, \quad V = 1 + y^2 - x^2.$$

We also put

$$S = x^2 + y^2 - 1.$$

Thus the notation is the same as in the contact test, now with the opposite-side length normalized to 1. The identities

$$U^2 + H^2 = 4x^2, \quad V^2 + H^2 = 4y^2, \quad S^2 + H^2 = 4x^2y^2 \quad (5)$$

will be used repeatedly.

Let  $\alpha \leq \beta \leq \gamma$  be the angles opposite  $x, y, 1$ . The four regimes below have a direct geometric meaning. The first boundary is  $\beta = 45^\circ$ , the second is  $\alpha = 45^\circ$ , and the third says that the altitude to the shortest side equals that side. Figure 3 shows the resulting shape diagram.

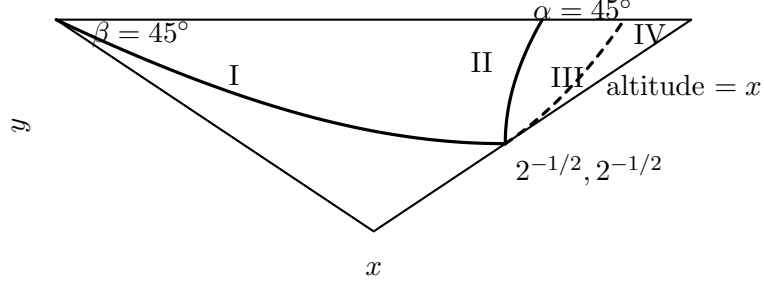


Figure 3: The normalized shape region  $0 < x \leq y \leq 1$ ,  $x + y > 1$ . Regions I–IV correspond to the four formulas in Theorem 6. The dashed boundary separates the side-square and vertex-square regimes.

**Theorem 6** (Smallest square). *For  $0 < x \leq y \leq 1$  and  $x + y > 1$ ,*

$$\rho(x, y) = \begin{cases} \frac{2x}{U}, & H \leq U, \\ \frac{2\sqrt{1+y^2-H}}{V}, & U \leq H \leq V, \\ \frac{2x}{H}, & V \leq H \text{ and } H \geq 2x^2, \\ \frac{2\sqrt{x^2+y^2-H}}{S}, & V \leq H \leq 2x^2. \end{cases} \quad (6)$$

*At common boundaries the neighboring formulas agree; the right-isosceles point is read by continuity.*

*Proof.* Because

$$\tan \beta = \frac{H}{U}, \quad \tan \alpha = \frac{H}{V},$$

the first two boundaries are exactly  $\beta = 45^\circ$  and  $\alpha = 45^\circ$ .

The only regime in which three-side contact differs from ordinary minimum-square containment is Region I. There  $\beta \leq 45^\circ$ . Use the longest side 1 as the opposite-side length. In the normalized form (4), the three upper bounds on  $t$  are

$$\frac{V}{H}, \quad \frac{H}{U}, \quad \frac{2-H}{V}.$$

Since  $H \leq U \leq V$ , the middle one is the smallest. Hence

$$t_{\max} = \frac{H}{U},$$

and by (5)

$$\rho = \sqrt{1 + t_{\max}^2} = \frac{2x}{U}.$$

Contacts based on a shorter side cannot improve this value in the interior of Region I: the compatibility condition for such a contact reduces to  $V \leq H$  or  $U \leq H$ , respectively, and these fail there. This proves the first line of (6).

In Regions II–IV the smallest containing square from the classical classification of Jerrard and Wetzel [2, Theorem 8] already has three-side contact. Indeed, the relevant minimizing square is either a vertex square, or a side square in which the perpendicular from the opposite vertex meets the supporting side segment. In the latter case the triangle can be slid parallel to that side until an endpoint reaches a corner without losing containment. Thus the classical minimum is also our minimum.

For Region II ( $\alpha \leq 45^\circ \leq \beta$ ), the vertex-square formula gives

$$m = \frac{V}{2\sqrt{1+y^2-H}},$$

which is the second line. If  $\alpha \geq 45^\circ$  and the altitude to the shortest side  $x$  is at least  $x$ , equivalently  $H \geq 2x^2$ , the minimizing square is a side square and

$$m = \frac{H}{2x},$$

giving the third line. In the remaining part of  $\alpha \geq 45^\circ$ , the minimizing square is a vertex square at the largest angle and

$$m = \frac{S}{2\sqrt{x^2+y^2-H}},$$

giving the fourth line. The formulas agree on the three separating curves.  $\square$

The theorem explains Figure 1. For  $(2, 2, 3)$  we have  $x = y = 2/3$ , which lies in Region I. Formula (6) gives  $\rho = 4/3$ , so

$$m(2, 2, 3) = \frac{3}{4/3} = \frac{9}{4}.$$

## 5 Integer-sided triangles

Every congruence class of integer-sided triangles has a unique ordered triple

$$1 \leq \ell_1 \leq \ell_2 \leq \ell_3,$$

with  $\ell_1 + \ell_2 > \ell_3$ . By Theorem 3, such a triangle is counted by  $a(n)$  exactly when

$$m(\ell_1, \ell_2, \ell_3) \leq n \leq \ell_3.$$

Moreover, some side joins opposite sides of the square, so necessarily  $\ell_3 \leq \sqrt{2}n$ .

**Corollary 7.** *The number  $a(n)$  equals the number of integer triples*

$$1 \leq \ell_1 \leq \ell_2 \leq \ell_3, \quad \ell_1 + \ell_2 > \ell_3,$$

for which

$$m(\ell_1, \ell_2, \ell_3) \leq n \leq \ell_3 \leq \sqrt{2}n.$$

This is a finite exact algorithm. One may evaluate  $m$  from Theorem 6, or use the six tests of Theorem 1 directly. Some values are

$$a(10) = 108, \quad a(20) = 738, \quad a(50) = 10448, \quad a(100) = 80606.$$

The two methods agree for all values checked through  $n = 100$ . Boundary equalities in the contact test can be handled exactly, so floating-point decisions are not needed.

The side joining opposite sides need not be the longest. The triangle  $(34, 34, 35)$  is counted for  $n = 33$ , but one of its sides of length 34 must join the opposite sides.

## 6 The growth of the counting function

The geometry also explains the order of growth of  $a(n)$  without further case analysis. Let  $\mathcal{R}$  be the set of real triples

$$0 < u \leq v \leq w, \quad u + v > w,$$

that have three-side contact with the unit square. Scaling gives

$$a(n) = \#(n\mathcal{R} \cap \mathbb{Z}^3). \quad (7)$$

The region  $\mathcal{R}$  is bounded, with  $1 \leq w \leq \sqrt{2}$ , and is semialgebraic by Theorem 1. Davenport's Lipschitz principle [1] therefore gives the following.

**Theorem 8.** *There is a positive constant  $C$  such that*

$$a(n) = Cn^3 + O(n^2),$$

and  $C = \text{vol}(\mathcal{R})$ .

The constant reduces to a two-dimensional integral. Normalize the longest side to 1. The shape region is

$$D = \{(x, y) : 1/2 < y \leq 1, 1 - y < x \leq y\}.$$

For a fixed shape, the scale variable  $z$  ranges from 1 to  $\rho(x, y)$ . The change of variables

$$(u, v, w) = (zx, zy, z)$$

has Jacobian  $z^2$ . Hence

$$C = \frac{1}{3} \int_{1/2}^1 \int_{1-y}^y (\rho(x, y)^3 - 1) dx dy. \quad (8)$$

Using the four formulas in Theorem 6 gives

$$C = 0.0776924549 \dots$$

## 7 Discussion

The three-side condition changes the classical minimum-square problem in one precise place: the diagonal-square regime. Outside that regime, the smallest containing square already has the required boundary contacts. Inside it, the contact condition forces a larger square, and the first line of (6) supplies the replacement formula.

Theorem 3 reduces the admissible square sizes of a fixed triangle to one interval. After normalizing the longest side, its lower endpoint is governed by three geometric transitions: two  $45^\circ$  angle thresholds and one altitude threshold.

For integer-sided triangles this turns a continuous placement problem into a finite arithmetic count and leads immediately to cubic growth. Several variants remain open. One may replace the square by a rectangle, require the three vertices to lie in the relative interiors of their assigned sides, or ask whether the constant in (8) has a simpler closed form.

## Use of artificial intelligence

OpenAI ChatGPT (GPT-5.6 Sol) was used for exploratory computation, algebraic and code checks, literature searches, and language editing. The author verified all mathematical statements and proofs and is responsible for the final content.

## References

- [1] H. Davenport, *On a principle of Lipschitz*, J. London Math. Soc. **26** (1951), 179–183. doi:10.1112/jlms/s1-26.3.179.
- [2] R. P. Jerrard and J. E. Wetzel, *Triangles in squares*, Elem. Math. **58** (2003), 65–72. doi:10.1007/S000170300009.
- [3] L. Yuan and R. Ding, *Triangles in squares*, Math. Gaz. **88** (2004), 219–225. doi:10.1017/S0025557200174947.